

Counting

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 1 AC: 0 Score: 100.00

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Codes

AI Code Tips

Description

In a directed connected graph G , there are N nodes and M edges. Among the M edges, for arbitrary given node k (we named the N nodes $1, 2, \dots, N$), if there are s edges point to it, we call the node k with a in-degree s , if there are t edges leads from it, we call the node k with a out-degree t . For the given graph G , we define the spanning tree as follows:

1. The tree contains N nodes and $N-1$ edges in G ;
2. The N nodes are connected.

For the above definition of the spanning tree, if there is a node v meets that, the in-degree of node v is 0, and the in-degree of all the remaining nodes is 1, we call this kind of tree as outward tree roots with v . Now the given task is following:

In arbitrary given connected graph G with N nodes and M edges, can you tell me how many different kinds of trees there are in G meet the following condition?

1. Maybe the tree is a outward tree roots with v , and also maybe it's only a spanning tree;
2. The tree must contains the given h edges, you should assume the h edged belong to G ;

You may assume that there is no same direction edge between any two nodes.

Input

In the first line, there is a single integer T , which indicates the test case. For each test case, the first line contains three integers N ($2 \leq N \leq 100$), M ($N-1 \leq M < 10000$), R ($0 \leq R \leq N$), which N is the number of node, M is the number of directed edge, R is the root of the outward tree you should count, if $R = 0$, you task is counting the number of the spanning tree. In the next M lines, each line contains two integer numbers x and y , which is one edge that node x points to node y . After that, a integer h is given, which indicates the edges that the trees you are counting must contain, then next h lines is following, each line contains two integers x and y , indicates one edge which leads from x to y .

Output

For each test case, the first line you should output is "Case #:", where the '#' is the id of the test case, and then you should output one integer in

the second line, which is the number of the outward tree roots witch R or the spanning tree (if $R = 0$) you have figured out, of course the tree must contains the given h edges. If the result is not less then 10007, please output the result mod 10007.

Samples

input	Copy
<pre>2 4 5 0 1 2 1 4 4 2 4 3 3 2 0 4 6 1 1 2 1 4 1 3 2 4 4 3 2 3 1 4 3</pre>	
output	Copy
<pre>Case 1: 8 Case 2: 2</pre>	

Source

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HZNUOJ is based on [HUSTOJ](#)

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